TEXTBOOK EVALUATION

I. Choose the correct answer.

1. $H_{2(g)} + Cl_{29(g)} \rightarrow 2HCl_{(g)}$ is a
   a. Decomposition Reaction
   b. Combination Reaction
   c. Single Displacement Reaction
   d. Double Displacement Reaction

2. Photolysis is a decomposition reaction caused by _____
   a. heat b. electricity
   c. light d. mechanical energy

3. A reaction between carbon and oxygen is represented by $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ + Heat. In which of the type(s), the above reaction can be classified?
   (i) Combination Reaction
   (ii) Combustion Reaction
   (iii) Decomposition Reaction
   (iv) Irreversible Reaction
   a. i and ii b. i and iv
   c. i, ii and iii d. i, ii and iv

4. The chemical equation
   $Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow BaSO_{4(s)}\downarrow + 2NaCl_{(aq)}$
   represents which of the following types of reaction?
   a. Neutralisation
   b. Combustion
   c. Precipitation
   d. Single displacement

5. Which of the following statements are correct about a chemical equilibrium?
   (i) It is dynamic in nature
   (ii) The rate of the forward and backward reactions are equal at equilibrium
   (iii) Irreversible reactions do not attain chemical equilibrium
   (iv) The concentration of reactants and products may be different
   a. i, ii and iii b. i, ii and iv
   c. ii, iii and iv d. i, iii and iv

6. A single displacement reaction is represented by $X_{(s)} + 2HCl_{(aq)} \rightarrow XCl_{2(aq)} + H_{2(g)}$. Which of the following(s) could be X.
   (i) Zn (ii) Ag (iii) Cu (iv) Mg.
   Choose the best pair.
   a. i and ii b. ii and iii
   c. iii and iv d. i and iv

7. Which of the following is not an "element + element → compound" type reaction?
   a. $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$
   b. $2K_{(s)} + Br_{2(l)} \rightarrow 2KBr_{(s)}$
   c. $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$
   d. $4Fe_{(s)} + 3O_{2(g)} \rightarrow 2Fe_2O_{3(s)}$

8. Which of the following represents a precipitation reaction?
   a. $A_{(s)} + B_{(s)} \rightarrow C_{(s)} + D_{(s)}$
   b. $A_{(s)} + B_{(aq)} \rightarrow C_{(aq)} + D_{(l)}$
   c. $A_{(aq)} + B_{(aq)} \rightarrow C_{(s)} + D_{(aq)}$
   d. $A_{(aq)} + B_{(s)} \rightarrow C_{(aq)} + D_{(l)}$

9. The pH of a solution is 3. Its $[OH^-]$ concentration is
   a. 1×10^{-3} M
   b. 3 M
   c. 1×10^{-11} M
   d. 11 M

10. Powdered $CaCO_3$ reacts more rapidly than flaky $CaCO_3$ because of _____.
    a. large surface area
    b. high pressure
    c. high concentration
    d. high temperature

II. Fill in the blanks

1. A reaction between an acid and a base is called _____.
2. When lithium metal is placed in hydrochloric acid, _____ gas is evolved.

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